

Numerical methods

Error: $e = a - \bar{a}$; $\bar{a}$ = approximate value

Types of error:

i) Basis of quantity:

a) Absolute error: $e = |a - \bar{a}|$

b) Relative error: $e = \frac{|a - \bar{a}|}{a}$

c) Percentage error: $e = \frac{|a - \bar{a}|}{a} \times 100\%$

ii) Basis of source:

a) Truncation error: Arises due to discarding of the terms after certain term. For eg:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \infty$$

$$e^1 = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots \infty$$

$$\text{We take, } e = 1 + 1 + \frac{1}{2!}$$

$$\text{Error} = \frac{1}{3!} + \frac{1}{4!} + \dots + \infty$$

b) Round off error: Produced due to rounding of number

For eg: If we have, $a = 5.9327$, rounding up to 2 d.p

$$\bar{a} = 5.93$$

$$\text{error} = |a - \bar{a}| = 5.9327 - 5.93 = 0.0027$$

c) Error in original data: error in input data

d) Propagated error: Error produced as the process goes on

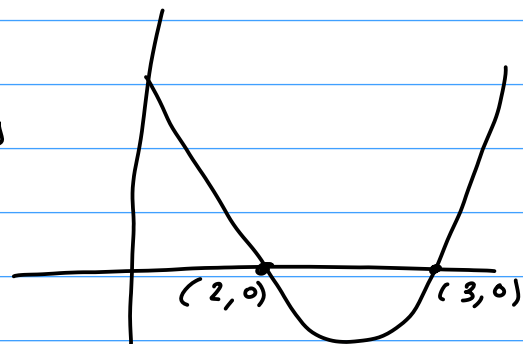
e) Blunder: error produced due to human imperfection.

Geometrical interpretation of roots:

Let $y = x^2 - 5x + 6$

x	2	3	0
y	0	0	6

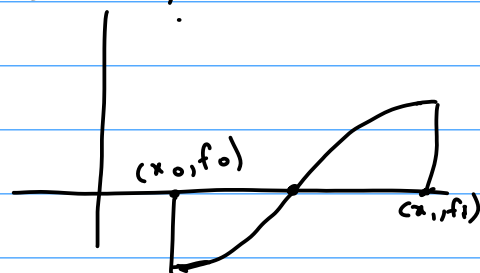
→ Interpretation: The points at which the function either cuts or touches the x-axis.



Bisection method:

Let $f(x) = 0$, Suppose two points x_0 and x_1 such that $f(x_0) \times f(x_1) < 0$

$$x_2 = \frac{x_0 + x_1}{2}$$



1) $e^x - x - 2 = 0$

x	0	1	1.5	1.1	1.2	1.15
f(x)	-1	-0.28	0.981	-0.0958	0.12	0.0081

Let $x_0 = 1.1$, $f_0 = -0.0958$

$x_1 = 1.15$, $f_1 = 0.0081$

$f(x_0) \times f(x_1) < 0$

$x_2 = \frac{x_0 + x_1}{2} = 1.125$; $f_2 = -0.0447$

$x_3 = \frac{x_2 + x_1}{2} = 1.1375$; $f_3 = -0.0185$

$x_4 = \frac{x_3 + x_1}{2} = 1.14375$; $f_4 = -0.0052$

$x_5 = \frac{x_4 + x_1}{2} = 1.146875$; $f_5 = 0.0014$

$x_6 = \frac{x_4 + x_5}{2} = 1.1453125$; $f_6 = -0.0018$

$$x_2 = \frac{x_6 + x_5}{2} = 1.14609375 ; f_2 = -0.0002$$

$$x_8 = \frac{x_7 + x_5}{2} = 1.146484375$$

$\therefore$ reqd root upto 3 d.p. is 1.146

Limitation of bisection:

$\rightarrow$ Determination of roots is guaranteed but the process is very slow.

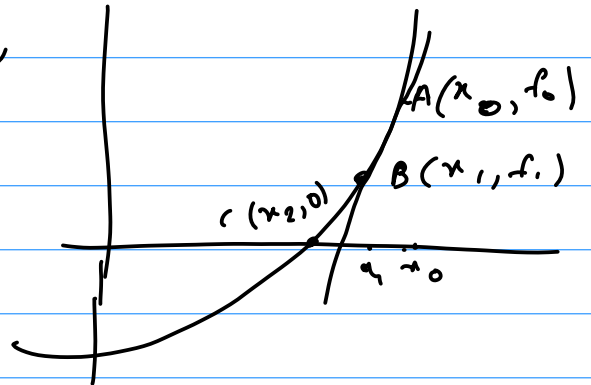
Secant method: (using secant)

Let $f(x) = 0$, choose two points x_0, x_1 near the root

We know, slope of AB = slope of BC,

$$\frac{f_1 - f_0}{x_1 - x_0} = \frac{0 - f_1}{x_2 - x_1}$$

$$x_2 - x_1 = -\frac{f_1 (x_1 - x_0)}{f_1 - f_0}$$



$$\text{OR, } x_2 = x_1 - \frac{f_1 (x_1 - x_0)}{f_1 - f_0}$$

$$\text{General formula: } x_{n+1} = x_n - \frac{f_n (x_n - x_{n-1})}{f_n - f_{n-1}}$$

Q) $\log x - \cos x$ upto 4 d.p

Let, $f(x) = \log(x) - \cos(x)$

x	1	1.5	1.2	1.3	1.31
$f(x)$	-0.54	0.334	-0.11	-0.0051	0.0121

$$x_0 = 1.3 ; f_0 = -0.0051$$

$$x_1 = 1.31 ; f_1 = 0.0121$$

$$x_2 = x_1 - \frac{f_1 (x_1 - x_0)}{f_1 - f_0} = 1.302965 ; f_2 = -0.000001$$

$$x_3 = x_2 - \frac{f_2 (x_2 - x_1)}{f_2 - f_1} = 1.302965$$

$\therefore$ reqd root is 1.3029

Limitations of secant method:

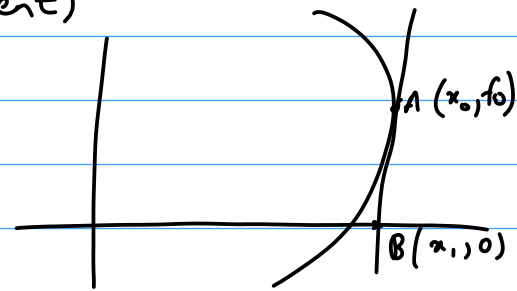
→ If slope is very small or zero, at that condⁿ root may not be determined

Newton-Raphson method: (using tangent)

→ Let $f(x) = 0$, choose x_0

Acc to coordinate system,

$$\text{Slope of AB} = \frac{0 - f_0}{x_1 - x_0} \rightarrow (1)$$



Acc to calculus,

$$\text{Slope of MB (m)} = f'(x_0) \rightarrow (2)$$

$$\text{From (1) \& (2), } \frac{-f_0}{x_1 - x_0} = f'(x_0)$$

$$\frac{-f_0}{f'_0} = x_1 - x_0$$

$$x_1 = x_0 - \frac{f_0}{f'_0}$$

General term,

$$x_{n+1} = x_n - \frac{f_n}{f'_n}$$

Q) $x - 1.5 \sin x - 2.5$ upto 4 d.p

x	2	2.5	2.7	2.8	2.9
f(x)	-1.86	-0.89	-0.44	-0.20	0.0411

$$x_0 = 2.9, f_0 = 0.041126$$

$$f'(x) = 1 - 1.5 \cos x, f'_0 = 2.456437$$

$$x_1 = x_0 - \frac{f_0}{f'_0} = 2.883257$$

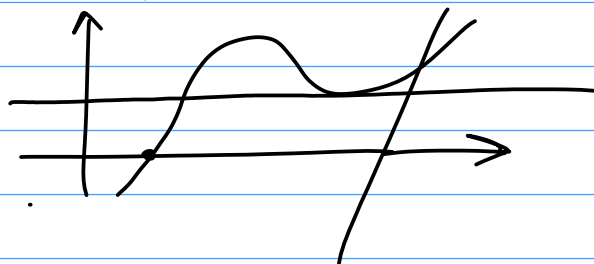
$$f_1 = 0.000041, f'_1 = 2.450224$$

$$x_2 = x_1 - \frac{f_1}{f'_1} = 2.883240$$

∴ The reqd root upto 4 d.p is 2.8832

Limitations:

- i) If f_n is too difficult for finding derivative, then this method is not useful
- ii) If value of derivative is very small or zero, root may not be determined.
- iii) If initial guessing is far from root, root cannot be determined



Fixed point iterative method,
 $f(x) = 0$, we convert it in the form of $x = g(x)$
then, $x_{n+1} = g(x_n)$

Q) $3x + \sin x - e^x = 0$, $x_0 = 0.36$

$$x = \frac{e^x - \sin x}{3} \quad ; \quad g(x) = \frac{e^x - \sin x}{3}$$

$$x_1 = \frac{e^{x_0} - \sin x_0}{3} = 0.360351$$

$$x_2 = \frac{e^{x_1} - \sin x_1}{3} = 0.360409$$

$$x_3 = \frac{e^{x_2} - \sin x_2}{3} = 0.360419$$

$\therefore$ The reqd root up to 4 dp is 0.3604

Convergence of fixed point iterative method

solⁿ let $f(x) = 0 \rightarrow$ ① be an eqn. We change ① in the form of
 $x = g(x) \rightarrow$ ②

$$\therefore x_{n+1} = g(x_n) \rightarrow$$
 ③

If p is a fixed point of $g(x)$ then, $p = g(p)$ ④

From ③ & ④,

$$x_{n+1} - p = g(x_n) - g(p) \rightarrow$$
 ⑤

using lagrange mean value theorem on R^n , of ⑤

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$
$$(b - a) f'(c) = f(b) - f(a)$$

or, $(x_n - p) g'(c) = g(x_n) - g(p)$

$$x_n, x_{n+1} - p = (x_n - p) g'(c)$$

$$|x_{n+1} - p| = |x_n - p| |g'(c)|$$

$$e_{n+1} = e_n k \rightarrow \textcircled{6} \text{ where } k \text{ is constant.}$$

$$e_{n+1} = e_{n-1} k \cdot k$$

$$e_{n+1} = e_{n-1} k^2 \rightarrow \textcircled{7}$$

$$e_{n+1} = e_{n-2} k^3$$

Proceeding further, $e_{n+1} = e_0 k^{n+1}$

$$\lim_{n \rightarrow \infty} e_{n+1} = \lim_{n \rightarrow \infty} e_0 k^{n+1}$$

$$\lim_{n \rightarrow \infty} |x_{n+1} - p| = 0 \quad \text{if } k < 1$$

$$\lim_{n \rightarrow \infty} x_{n+1} = p \quad \text{if } k < 1$$

$$\lim_{n \rightarrow \infty} x_{n+1} = p \quad \text{if } |g'(c)| < 1$$

$$\lim_{n \rightarrow \infty} x_{n+1} = p \quad \text{if } |g'(x_0)| < 1$$

$\therefore$ Fixed point iterative method is convergent if $|g'(x_0)| < 1$

Convergence of Newton Raphson's method:

$$\text{Let } f(x) = 0 \rightarrow \textcircled{1}$$

$$\text{By Newton Raphson method, } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \rightarrow \textcircled{2}$$

Again, from fixed point iterative method,

$$x_{n+1} = g(x_n) \rightarrow \textcircled{3}$$

From $\textcircled{2} \wedge \textcircled{3}$,

$$g(x_n) = x_n - \frac{f(x_n)}{f'(x_n)} \rightarrow \textcircled{4}$$

$$g(x) = x - \frac{f(x)}{f'(x)} \rightarrow \textcircled{5}$$

D.b.s. w.r.t. x ,

$$g'(x) = 1 - \frac{(f'(x) \cdot f'(x) - f(x) \cdot f''(x))}{[f'(x)]^2}$$

$$g'(x) = \frac{[f'(x)]^2 - [f'(x)]^2 + f(x) \cdot f''(x)}{[f'(x)]^2}$$

$$g'(x) = \frac{f(x) \cdot f''(x)}{[f'(x)]^2}$$

$$\left| \frac{f(x) \cdot f''(x)}{[f'(x)]^2} \right| = |g'(x)|$$

The process is convergent if $|g'(x_0)| < 1$
" " " " " $\left| \frac{f(x_0) \cdot f''(x_0)}{[f'(x_0)]^2} \right| < 1$

$\therefore$ Newton Raphson Method is convergent if $\left| \frac{f(x_0) \cdot f''(x_0)}{[f'(x_0)]^2} \right| < 1$ ⑥

It may be tedious to check the condⁿ ⑥, so to check the convergence of any method. We simply check the condⁿ
 $|x_{n+1} - x_n| > |x_n - x_{n-1}| \rightarrow \text{⑦}$

if ⑦ is true, at that condⁿ convergence is assured

Interpolation:

Short note: on interpolation and extrapolation,

→ Interpolation is a mathematical technique in which value of a function in the interval (x_0, x_n) is determined by determining the function at first is called interpolation.

While the determination of the function outside the interval is called extrapolation.

$$\text{Eg: } \begin{array}{c|c|c|c} x & 0 & 3 & 6 \\ \hline f(x) & 20 & 14 & 26 \end{array}$$

$$f(1) = ?$$

Consider a fn, $f(x) = a_0 + a_1x + a_2x^2 \rightarrow \textcircled{1}$

$$f(0) = a_0$$

$$a_0 = 20$$

$$\begin{array}{l|l} f(3) = a_0 + 3a_1 + 9a_2 & f(6) = a_0 + 6a_1 + 36a_2 \\ 3a_1 + 9a_2 = -6 & 6 = 6a_1 + 36a_2 \\ a_1 + 3a_2 = -2 \rightarrow \textcircled{2} & a_1 + 6a_2 = 1 \rightarrow \textcircled{3} \end{array}$$

Solving $\textcircled{2}$ & $\textcircled{3}$, $a_1 = -5$, $a_2 = 1$

Now $\textcircled{1}$ is, $f(x) = 20 - 5x + x^2$

$$\therefore f(1) = 20 - 5 + 1 = 16 //$$

Lagrange interpolating polynomial:

→ For 3rd degree polynomial,

$$f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}f_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}f_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}f_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}f_3$$

Q) Find cubic polynomial . Find $f(1.5)$

x	0	1	2	3
$f(x)$	1	2	1	10

Solⁿ,

We have Lagrange^{interpolating} polynomial

$$f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}f_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}f_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}f_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}f_3$$

$$\therefore f(x) = \frac{(x-1)(x-2)(x-3)}{(-1)(-2)(-3)}f_0 + \frac{(x-0)(x-2)(x-3)}{(1)(-1)(-2)}f_1 + \frac{(x-0)(x-1)(x-3)}{(2)(1)(-1)}f_2 + \frac{(x-0)(x-1)(x-2)}{(3)(2)(1)}f_3$$

$$f(x) = \frac{(x-1)(x-2)(x-3)}{-6} + \frac{x(x-2)(x-3)}{2} - \frac{x(x-1)(x-3)}{6} + \frac{10x(x-1)(x-2)}{6}$$

$$\therefore f(1.5) = 1 //$$

Newton's interpolating polynomial by divided difference method.

$$f(x) = f_0 + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) + f[x_0, x_1, x_2, x_3](x-x_0)(x-x_1)(x-x_2) + \dots$$

$$\text{where, } f[x_0, x_1] = \frac{f_1 - f_0}{x_1 - x_0}$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_1}$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$$

$$f[x_0, x_1, x_2, \dots, x_n] = \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0}$$

Q) $f(35) = ?$

x	25	30	40	50
$f(x)$	52	67.3	84.1	94.4

Using Newton's divide and difference method

The reqd table is

	x	f_i	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, x_{i+1}, x_{i+2}, x_{i+3}]$
$x_0 =$	25	52	$f[x_0, x_1] = 3.06$	$f[x_0, x_1, x_2] = -0.092$	$f[x_0, x_1, x_2, x_3] = 0.00238$
$x_1 =$	30	67.3	$f[x_1, x_2] = 1.68$	$f[x_1, x_2, x_3] = -0.0325$	
$x_2 =$	40	84.1	$f[x_2, x_3] = 1.03$		
$x_3 =$	50	94.4			

$$f(x) = f_0 + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2)$$

$$\therefore f(x) = 52 + 3.06(x - 25) - 0.092(x - 25)(x - 30) + 0.00238(x - 25)(x - 30)(x - 40)$$

$$\therefore f(35) = 77.405 //$$

Newton's forward interpolating polynomial. [near the beginning of observation]

$$f(x) = f_0 + s \Delta f_0 + \frac{s(s-1)}{2!} \Delta^2 f_0 + \frac{s(s-1)(s-2)}{3!} \Delta^3 f_0 + \frac{s(s-1)(s-2)(s-3)}{4!} \Delta^4 f_0 + \dots$$

where $h = x_{i+1} - x_i$ for all i ,

$$h = x_1 - x_0 = x_2 - x_1 \dots$$

$$s = \frac{x - x_0}{h};$$

$$\Delta f_0 = f_1 - f_0$$

$$\Delta f_n = f_{n+1} - f_n$$

$$\Delta f_1 = f_2 - f_1$$

$$\Delta^2 f_0 = \Delta \cdot \Delta f_0 = \Delta(f_1 - f_0) = \Delta f_1 - \Delta f_0$$

$$\Rightarrow f_2 - f_1 - (f_1 - f_0) = f_2 - 2f_1 + f_0$$

Q) Find NFIP. find $f(1.35)$

x	1.2	1.3	1.4	1.5
$f(x)$	1.063	1.091	1.119	1.145

Sol,

The NFIP table is

	x	f_i	Δf_i	$\Delta^2 f_i$	$\Delta^3 f_i$
x_0	1.2	1.063	$\Delta f_0 = 0.028$	$\Delta^2 f_0 = 0$	$\Delta^3 f_0 = -0.002$
x_1	1.3	1.091	$\Delta f_1 = 0.028$	$\Delta^2 f_1 = -0.002$	
x_2	1.4	1.119	$\Delta f_2 = 0.026$		
x_3	1.5	1.145			

$$h = x_1 - x_0 = 0.1, \quad S = \frac{x - x_0}{h} = \frac{x - 1.2}{0.1}$$

$$S = 10x - 12$$

$$f(x) = f_0 + S \Delta f_0 + \frac{S(S-1)}{2!} \Delta^2 f_0 + \frac{S(S-1)(S-2)}{3!} \Delta^3 f_0$$

$$f(x) = 1.063 + (10x - 12)0.028 + \frac{1}{2} \left\{ (10x - 12)(10x - 13) \right\} \times 0.002 + \frac{(10x - 12)(10x - 13)(10x - 14)}{3!}$$

$$f(1.35) = 1.105125 //$$

Newton's backward interpolating polynomial (near end of observation)

$$f(x) = f_n + p \nabla f_n + \frac{p(p+1)}{2!} \nabla^2 f_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 f_n + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 f_n + \dots$$

$$h = x_{i+1} - x_i = x_1 - x_0 = \dots$$

$$p = \frac{x - x_n}{h}$$

$$f_n = f_n - f_{n-1}$$

Q) Find $f(60-70)$

x	0-40	40-60	60-80	80-100	100-120
$f(x)$	250	120	100	70	50

Solⁿ, Rearranging the given obs as,

x	f_i	∇f_i	$\nabla^2 f_i$	$\nabla^3 f_i$	$\nabla^4 f_i$
40	250				
60	370	$\nabla f_1 = 120$			
80	470	$\nabla f_2 = 100$	$\nabla^2 f_1 = -20$		
100	540	$\nabla f_3 = 70$	$\nabla^2 f_2 = -30$	$\nabla^3 f_3 = -10$	
120	590	$\nabla f_4 = 50$	$\nabla^2 f_3 = -20$	$\nabla^3 f_4 = 10$	$\nabla^4 f_4 = 20$

$$h = 20, \quad p = \frac{x - 120}{20} = \frac{x}{20} - 6 = 0.05x - 6$$

$$f(x) = 590 + (0.05x-6)50 - \frac{(0.05x-6)(0.05-5)20}{2} + \frac{(0.05x-6)(0.05-5)(0.05-4)10}{3!} \\ + \frac{(0.05x-6)(0.05-5)(0.05-4)(0.05-3)20}{4!}$$

$$f(70) = 423.59 \approx 424$$

$$\therefore f(60-70) = f(70) - f(60) \\ = 424 - 370 \\ = \underline{\underline{54}}$$

Fitting of curve:

- Polynomial is determined according to nature of given observation.
- More accurate than interpolation.

Fitting of straight curve:

We have curve, $y = ax + b \rightarrow \textcircled{1}$

The condⁿ for best fitting are,

$$\sum y_i = a \sum x_i + bn \rightarrow \textcircled{2}$$

$$\sum y_i x_i = a \sum x_i^2 + b \sum x_i \rightarrow \textcircled{3}$$

The reqd table is,

Put values of variables in $\textcircled{2}$ & $\textcircled{3}$

Solve $\textcircled{2}$ & $\textcircled{3}$ to get values of a and b ,

Put a and b in $\textcircled{1}$

$\therefore$ That is the best fitting curve.

Fitting of quadratic (parabolic) (2^{nd} degree curve).

$$y = ax^2 + bx + c \rightarrow \textcircled{1}$$

Condⁿ for best fitting curve are,

$$\sum y_i = a \sum x_i^2 + b \sum x_i + cn \rightarrow \textcircled{2}$$

$$\sum y_i x_i = a \sum x_i^3 + b \sum x_i^2 + c \sum x_i \rightarrow \textcircled{3}$$

$$\sum y_i x_i^2 = a \sum x_i^4 + b \sum x_i^3 + c \sum x_i^2 \rightarrow \textcircled{4}$$

On solving $\textcircled{2}$, $\textcircled{3}$, $\textcircled{4}$ we get a, b, c .

On putting values in $\textcircled{1}$, we get best fitting curve.

Fitting of transcendental function.

→ The relation betⁿ dependent & independent variable may not be always in polynomial form.

Q) Fit curve of form, $y = ae^{bx}$ to the following observations.

x	2	4	6	8	10
y	4.027	11.084	30.128	81.897	222.62

Solⁿ,

Given model is $y = ae^{bx} \rightarrow (1)$

$$\ln y = \ln(ae^{bx})$$

$$\ln y = \ln(a) + bx \ln(e)$$

$$\ln y = \ln a + bx$$

Condⁿ for best fitting are,

$$\sum \ln y_i = n \ln a + b \sum x_i \rightarrow (2)$$

$$\sum \ln y_i \cdot x_i = \ln a \sum x_i + b \sum x_i^2 \rightarrow (3)$$

The req^d table is,

x	y	x^2	$\ln y$	$x \ln y$
2	4.027			
4	11.084			
6	30.128			
8	81.897			
10	222.62			

} Calculate the values

$$\sum x_i = 30 \quad \sum x_i^2 = 220 \quad \sum \ln y_i = 17.025 \quad \sum \ln y_i \cdot x_i = 122.15$$

$$17.025 = 5 \ln a + 30b \rightarrow (4)$$

$$122.15 = 30 \ln a + 220b \rightarrow (5)$$

Solving (4) and (5),

$$\ln a = 0.405$$

$$b = 0.5$$

$$e^{\ln a} = a = a = e^{0.405} = 1.499 \approx 1.5$$

Now, put in (1)

$y = 1.5 e^{0.5x}$ is the best fitting function.

Q) Fit a curve of form $PV^r = \beta \rightarrow (1)$

P	:	:	:	:
V	:	:	:	:

Taking log on both sides,

$$\log(PV^r) = \log(\beta)$$

$$\log P + r \log(V) = \log \beta$$

$$\log P = \log \beta - r \log(V)$$

Best fitting condⁿ are,

$$\sum \log P_i = n \log \beta - r \sum \log V_i \rightarrow (2)$$

$$\sum \log P_i \cdot \log V_i = \log \beta \sum \log V_i - r \sum (\log V_i)^2 \rightarrow (3)$$

Q) Fit the model of the form $T = at + be^{t/4} \rightarrow (1)$

t	-	-	-
T	-	-	-

Best fitting condⁿ are,

$$\sum T_i = na + b \sum e^{t_i/4} \rightarrow (2)$$

$$\sum T_i \cdot e^{t_i/4} = a \sum e^{t_i/4} + b \sum (e^{t_i/4})^2 \rightarrow (3)$$

Q) Fitting a curve of type $y = \frac{1}{ax+b} \rightarrow (1)$

$$\text{or, } \frac{1}{y} = ax + b$$

Best fitting condⁿs are,

$$\sum \frac{1}{y_i} = a \sum x_i + nb \rightarrow (2)$$

$$\sum \frac{x_i}{y_i} = a \sum x_i^2 + b \sum x_i \rightarrow (3)$$

Q) $y = a + bx^2 \rightarrow (1)$

$$\sum y_i = na + b \sum x_i^2 \rightarrow (2)$$

$$\sum y_i x_i^2 = a \sum x_i^2 + b \sum (x_i)^4 \rightarrow (3)$$

Q) $y = ax + bx^2 \rightarrow (1)$

$$\text{or, } \frac{y}{x} = a + bx$$

Now, $\sum \frac{y_i}{x_i} = na + b \sum x_i \rightarrow (2)$

$$\sum \frac{y_i}{x_i} \cdot x_i = a \sum x_i + b \sum x_i^2 \text{ or, } \sum y_i = a \sum x_i + b \sum x_i^2 \rightarrow (3)$$

$$Q) y = ax + \frac{b}{x} \rightarrow (1)$$

$$\text{or, } xy = ax^2 + b$$

$$\sum x_i y_i = a \sum x_i^2 + nb \rightarrow (2)$$

$$\sum x_i^3 y_i = a \sum x_i^4 + b \sum x_i^2 \rightarrow (3)$$

$$Q) y = ax^2 + \frac{b}{x} \rightarrow (1)$$

$$\text{or, } xy = ax^3 + b$$

$$\sum x_i y_i = a \sum x_i^3 + nb \rightarrow (2)$$

$$\sum x_i^4 y_i = a \sum x_i^6 + b \sum x_i^3 \rightarrow (3)$$

Numerical differentiation and integration.

Numerical differentiation: [Note: h = interval betⁿ observation]
→ From Taylor's series:
$$f(x+h) = f(x) + hf'(x) + \underbrace{\frac{h^2}{2!} f''(x) + \dots}_{\text{neglect}} \infty$$

$$f(x+h) = f(x) + hf'(x)$$

i) Forward difference quotient:
$$f(x+h) = f(x) + hf'(x) \rightarrow \textcircled{1}$$

$$\therefore f'(x) = \frac{f(x+h) - f(x)}{h}$$

ii) Backward difference quotient;
$$f(x-h) = f(x) - hf'(x) \rightarrow \textcircled{2}$$

$$\therefore f'(x) = \frac{f(x) - f(x-h)}{h}$$

iii) Central difference quotient:
Do $\textcircled{1} - \textcircled{2}$
$$f(x+h) - f(x-h) = 2hf'(x)$$

$$\therefore f'(x) = \frac{f(x+h) - f(x-h)}{2h}$$

Note: Central is more accurate than forward and backward

Q)

time(t)	5	6	7	8
distance f(t)	10	18.6	23.5	33.4

Calculate velocity at $t=6$

$$f'(6) = ?$$

Solⁿ

$$h = 6 - 5 = \underline{1}$$

$$f'(t) = \frac{f(t+h) - f(t-h)}{2h}$$

$$f'(6) = \frac{f(6+1) - f(6-1)}{2} = \frac{f(7) - f(5)}{2} = \frac{23.5 - 10}{2} = 6.75 \text{ m/s}$$

For velocity at $t=5$ sec, we need forward difference quotient.
 $f'(5) = \frac{f(5+1) - f(5)}{1} = f(6) - f(5) = 8.6 \text{ m/s}$

For velocity at $t=8$, we need backward difference quotient
 $f'(8) = \frac{f(8) - f(8-1)}{1} = f(8) - f(7) = 9.9 \text{ m/s}$

Velocity at $t=6.5$ sec,

→ it is near the beginning of observations, so we use Newton's forward interpolating polynomial.

x_i	f_i	Δf_i	$\Delta^2 f_i$	$\Delta^3 f_i$
5	10	$\Delta f_0 = 8.6$	$\Delta^2 f_0 = -3.7$	$\Delta^3 f_0 = 8.7$
6	18.6	$\Delta f_1 = 4.9$	$\Delta^2 f_1 = 5$	
7	23.5	$\Delta f_2 = 9.9$		
8	33.4			

$$h = 6 - 5 = 1, \quad s = \frac{x - 5}{1} = x - 5$$

NFIP is,

$$f(x) = f_0 + s \Delta f_0 + \frac{s(s-1)}{2!} \Delta^2 f_0 + \frac{s(s-1)(s-2)}{3!} \Delta^3 f_0$$

$$= 10 + (x-5)8.6 - \frac{(x-5)(x-6)3.7}{2!} + \frac{(x-5)(x-6)(x-7)8.7}{3!}$$

$$\therefore f(6.5) = 20.96$$

Now,

$$f'(t) = 8.6 - \frac{3.7}{2} \left\{ (t-5) + (t-6) \right\} + \frac{8.7}{6} \left\{ (t-5)(t-6) + (t-6)(t-7) + (t-5)(t-7) \right\}$$

$$\therefore f'(6.5) = 4.5375 \text{ m/s}$$

Second order derivative:

From Taylor series,

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x)$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2!} f''(x)$$

$$+ \quad + \quad + \quad +$$

$$f(x+h) + f(x-h) = 2f(x) + h^2 f''(x)$$

$$\therefore f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

Numerical integration:

i) Trapezoid integration:

$$I = \frac{h}{2} \left[f_0 + f_n + 2(f_1 + f_2 + f_3 + \dots + f_{n-1}) \right]$$

ii) Simpson's $1/3$ rule:

$$I = \frac{h}{3} \left[f_0 + f_n + 4(f_1 + f_3 + f_5 + \dots + f_{n-1}) + 2(f_2 + f_4 + f_6 + \dots + f_{n-2}) \right]$$

iii) Simpson's $3/8$ rule:

$$I = \frac{3h}{8} \left[f_0 + f_n + 2(f_3 + f_6 + f_9 + \dots) + 3(f_1 + f_2 + f_4 + f_5 + \dots) \right]$$

where $h = \frac{b-a}{n}$, where $b =$ upper limit,
 $a =$ lower limit,
 $I = \int_a^b f(x) dx$.

T	1/3	3/8
1	2	3

$$LCM = 6$$

$\therefore$ Minimum number of intervals (n)
for all 3 rules is 6

Q) Evaluate $\int_0^\pi \sin x dx$ by all 3 rules and compare each with exact value.
Soln, $x_0 = a = 0$, $b = \pi$ ($n = 6$)

$$h = \frac{\pi - 0}{n} = \frac{\pi}{6}; \quad x_i = x_{i-1} + h$$

i	0	1	2	3	4	5	6
x_i	0	$\pi/6$	$2\pi/6$	$3\pi/6$	$4\pi/6$	$5\pi/6$	π
f_i	0	$1/2$	$\sqrt{3}/2$	1	$\sqrt{3}/2$	$1/2$	0

$$\text{Exact integration} = \int_0^\pi \sin x dx = -[\cos x]_0^\pi = -(\cos \pi - \cos 0) = 2$$

i) Trapezoidal rule,

$$I = \frac{h}{2} [f_0 + f_6 + 2(f_1 + f_2 + f_3 + f_4 + f_5)] = 1.954$$

$$\text{Error} = \left| \frac{\text{Exact} - I}{\text{Exact}} \right| \times 100\% = 2.3\%$$

ii) Simpson's 1/3 rule:

$$I = \frac{h}{3} [f_0 + f_6 + 4(f_1 + f_3 + f_5) + 2(f_2 + f_4)] = 2.0008$$

$$\text{Error} = \left| \frac{\text{Exact} - I}{\text{Exact}} \right| \times 100\% = 0.04\%$$

iii) Simpson's 3/8 rule:

$$I = \frac{3h}{8} [f_0 + f_6 + 2(f_3) + 3(f_1 + f_2 + f_4 + f_5)] = 2.002$$

$$\text{Error} = \left| \frac{\text{Exact} - I}{\text{Exact}} \right| \times 100\% = 0.1\%$$